

APEX Semester Planning Agent

Project report

Live demo: <https://technion-semester-planner-agent.vercel.app>

Project Summary

APEX is a Technion semester-planning agent. The student describes their situation in natural language, and the system builds a semester plan with courses, timetable, exams, risks, and explanations.

The project has two parts. The offline part is a data pipeline that prepares reliable course data. The online part is an agent that chooses tools, checks plans, repairs them, and answers the student.

Report Scope

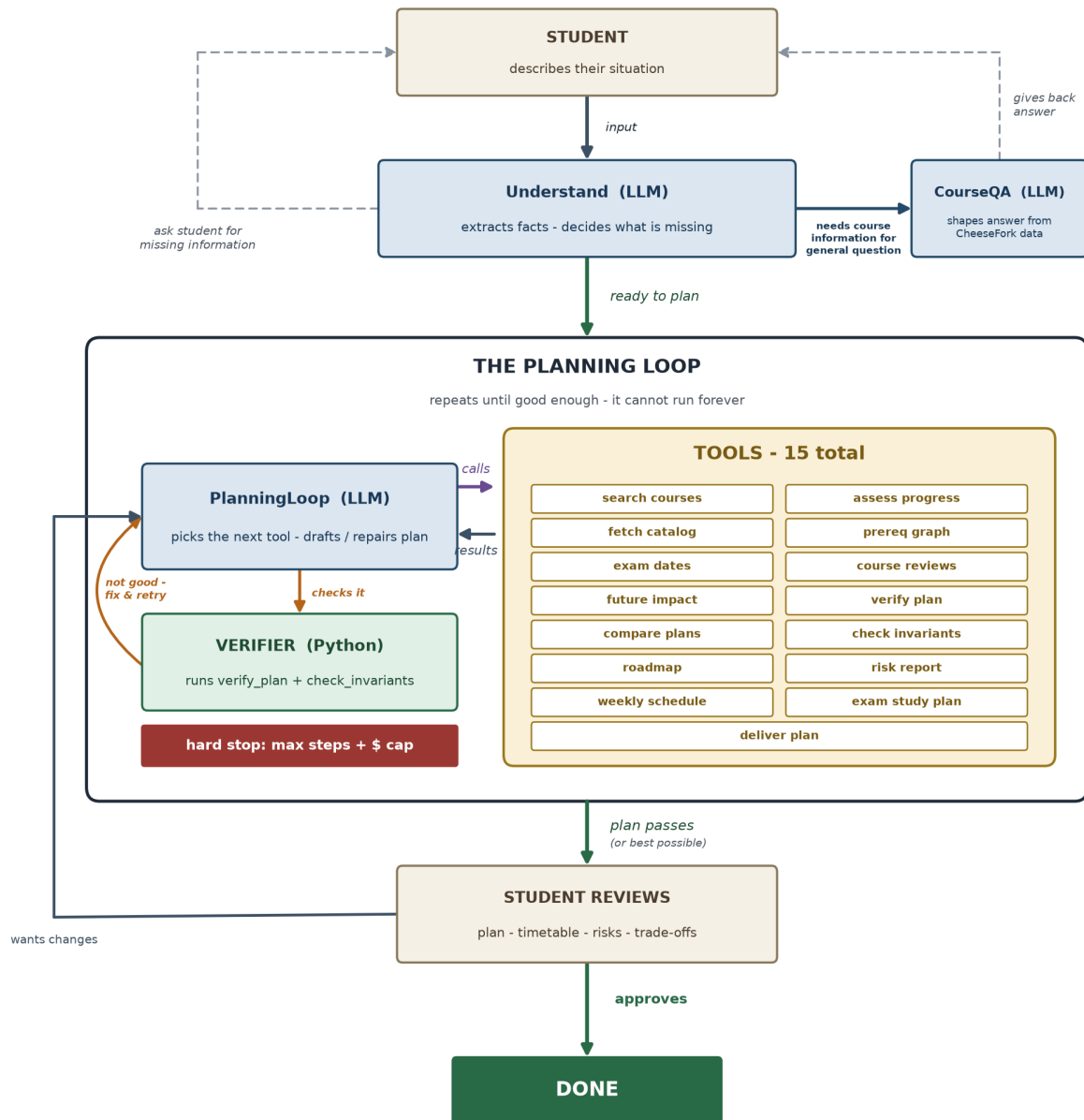
This report focuses on the system design: architecture, data, tools, endpoint contract, and known limitations. The companion document, *What APEX Can Do*, focuses on user-facing examples and screenshots from the working product.

Architecture

The default runtime is a bounded ReAct-style loop. The model does not produce a plan in one shot. It receives a set of tools, chooses which tool to call, reads the tool result, and decides the next step.

Module	Role
Understand	Extracts structured student state from the conversation.
PlanningLoop	Chooses tools, drafts plans, verifies plans, and repairs when needed.
CourseQA	Answers focused course questions without running a full planner.

Semester Planning Agent - Model Architecture



Tools

The agent has exactly 15 model-callable tools. Three diagnostics are also injected automatically every planning turn to reduce cost and avoid repeated calls: progress assessment, catalog context, and prerequisite graph context.

Tool area	Examples
Course search and catalog	<code>search_courses</code> , <code>fetch_catalog</code> , <code>query_prereq_graph</code>
Reviews and grades	<code>summarize_cheesefork</code> , grade-improvement analysis from technion-histograms
Plan checks	<code>verify_plan</code> , <code>check_invariants</code> , <code>compare_plans</code>
Schedule and exams	<code>weekly_schedule_analyzer</code> , <code>fetch_exam_dates</code> , <code>exam_study_planner</code>
Progress and risk	<code>assess_progress</code> , <code>simulate_future_impact</code> , <code>roadmap_to_graduation</code> , <code>risk_report</code>
Final answer	<code>deliver_plan</code>

Data Sources

Source	Used for
Technion course API	Catalog, requirements, prerequisites, schedules, sections, and exam dates.
Official track diagrams	Correct semester placement for mandatory courses.
CheeseFork	Student reviews, difficulty, satisfaction, and load.
technion-histograms	Historical course averages for grade-improvement reasoning.
Student input	Passed courses, failed courses, grades, and transcript PDF.
Supabase	Optional session persistence.

Limitations and Future Enhancements

The planner currently gives the model a compact but information-rich planning context for the current semester. This makes the agent strong at comparing courses, workload, reviews, and exam spacing in one pass, but it also makes the planning prompt longer than a minimal chat prompt.

The current design keeps core planning facts in structured tools because prerequisites, credits, section times, and exam dates need exact checks. A future version could add retrieval for policy

documents and administrative guidance, while keeping scheduling and degree-rule validation deterministic.

- The data bundle is a snapshot. It will not automatically update when Technion publishes a new catalog unless the offline pipeline is rerun.
- The system does not register students to courses.
- The system does not access private Technion accounts.
- The agent can report a best honest plan when constraints make a perfect plan impossible.
- Future work could add a small policy-retrieval layer for regulations, forms, and administrative instructions.

Required Endpoints

Endpoint	Purpose
GET /api/team_info	Returns the required team metadata for the assignment.
GET /api/agent_info	Returns description, purpose, prompt template, examples, and steps.
GET /api/model_architecture	Returns the architecture diagram as PNG.
POST /api/execute	Runs a one-shot prompt and returns <code>status</code> , <code>error</code> , <code>response</code> , and <code>steps</code> .